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Design Review: Reflection #2

Our team project, developed in collaboration with the Santa Clara University Access Office (AO), seeks to improve sustainability and operational efficiency by using data visualization to track food consumption trends across campus dining services. In focusing on food waste reduction and labor optimization, our project engages directly with civic issues such as environmental sustainability, ethical labor practices, and public transparency. These issues, while local in execution, are rooted in broader public policy frameworks and involve the oversight and participation of various governmental and non-governmental organizations.

Seeing as we are dealing with data provided to us by a third party much of the typical problems involved are handled by Transact. Compliance with CCPA and SOPIPA is not something that we are meant to handle and as long as we can ensure that Transact has the requisite certifications. We however must work hard to prevent data breaches or otherwise distribute the data in an improper way which could violate these laws. This can be done by following many common best practices in writing code as well as using NIST guidelines. Furthermore, if we ever choose to use any cloud provider they make their compliance and security measures known so we must make sure they meet all our standards. In addition, although not a direct responsibility, we also have a duty to the students of Santa Clara to improve the service for them.

At the heart of our project lies a key civic issue: food waste. Food waste represents a significant environmental challenge with both local and global implications. According to the U.S. Environmental Protection Agency (EPA), food waste

is the largest single category of material placed in municipal landfills, contributing to methane emissions and climate change (EPA, 2023). Our effort to reduce food waste at SCU aligns with EPA guidelines and California's SB 1383 law, which mandates reductions in organic waste disposal. Thus, our project operates within a regulatory context that includes state legislation designed to combat greenhouse gas emissions and promote composting and food recovery programs.

In addition, our project supports the university's own sustainability goals. Santa Clara University, like many institutions of higher education, participates in civic engagement through sustainability pledges, community wellness programs, and environmental education. Our system contributes to these goals by encouraging data driven decision making and promoting a culture of sustainable food consumption. Moreover, by making the data accessible and actionable for AO leadership, we support transparent governance practices within the university's operational structure.

Although our project is starting at a university, our solution placement can be extended to city-wide or county-level public institutions that can also manage dining services, such as public schools, correctional facilities, or hospitals. This expansion might raise concerns for different local government regulations. Departments such as the Santa Clara County Department of Environmental Health might have policies or guidelines that our product would need to be in compliance with.

In addition, our data might eventually help identify patterns that support SCU's contribution to regional climate action plans or national benchmarks such as those set by the United Nations' Sustainable Development Goals. Our product not only improves efficiency but it also plays a role in civic responsibility. From data privacy and regulatory compliance to environmental management and institutional transparency, our project shows how engineering and public service can come together.

Sources

California Legislative Information. (2016). *SB-1383 Short-lived climate pollutants: methane emissions: dairy and livestock: organic waste: landfills*.
<https://leginfo.legislature.ca.gov>

U.S. Environmental Protection Agency (EPA). (2023). *Food Loss and Waste*.
<https://www.epa.gov/sustainable-management-food/reducing-wasted-food-home>

